

1. Explain the difference between a Tableau Dashboard and a Tableau Story Point. Give two healthcare-related use cases.

Feature	Tableau Dashboard	Tableau Story Point
Definition	A collection of several worksheets (views) arranged on a single screen to compare and monitor data simultaneously.	A sequence of visualizations, dashboards, or worksheets that work together to convey information or a narrative.
Primary Purpose	Exploration & Monitoring: Designed for operational monitoring, real-time tracking, and interactive data filtering by the user.	Explanation & Presentation: Designed to guide an audience through a specific narrative arc, finding, or business case.
Interactivity	High. Users can toggle multiple global filters and actions to explore data deeply on their own terms.	Controlled. It preserves a specific snapshot or state of data for each step, though users can still interact within that specific frame.

Use Case A: Hospital Operational Dashboard

- **The Setup:** A live, single-screen view containing charts for *Current Emergency Room (ER) Wait Times*, *Available Bed Capacity by Ward*, and *On-Duty Staffing Levels*.
- **Why a Dashboard?** The Chief Operating Officer needs to monitor real-time resource constraints. They can click a filter to view only "ICU Beds" to instantly check capacity without needing a structured narrative.

Use Case B: Annual Patient Safety & Quality Improvement Story

- **The Setup:** A 5-point sequence tracking a historical trend:
 - i. *Point 1:* A line chart showing a sudden spike in post-surgical infections in Q2.
 - ii. *Point 2:* A breakdown highlighting that the spike occurred primarily in a specific operating wing.
 - iii. *Point 3:* A dashboard correlating the spike with a temporary shortage of specialized sterilization staff during those months.
 - iv. *Point 4:* A chart showing the recovery of metrics after new training protocols were implemented.
- **Why a Story Point?** The team needs to explain a complex past event, justify budget expenditures, and prove the efficacy of a solution to board members who require context, not just raw exploration.

2. What are Actions in Tableau? Explain how Filter Action can help hospital management analyze patient data.

- Actions in Tableau are interactive features that add context, depth, and connectivity to your data visualizations. Instead of static charts, actions allow user movements—such as **hovering, clicking (selecting), or right-clicking (menu)**—to trigger a dynamic response across other worksheets, dashboards, or external web links.

The primary types of actions include:

- a. **Filter Actions:** Uses data from one view to filter data in other views.
- b. **Highlight Actions:** Draws attention to specific marks or categories across sheets while dimming unrelated data.
- c. **URL Actions:** Opens external web pages or triggers external processes based on a clicked data point.
- d. **Parameter/Set Actions:** Dynamically changes parameter values or updates set memberships based on user interaction.

A Filter Action transforms a passive view into an analytical funnel. For hospital management tracking complex healthcare metrics, it acts as an intuitive, click-to-drill-down discovery tool.

Here is how a Filter Action drastically improves patient data analysis:

- **Granular Drill-Down Without Screen Clutter:** Management can view a high-level KPI dashboard showing the overall admission counts or total treatment costs. By simply clicking on a specific region (e.g., "North") or a specific department (e.g., "ICU"), the rest of the charts on the dashboard instantly recalculate to show data *only* for that selection.
- **Cross-Contextual Analysis:** If a hospital executive notices a sudden spike in "Critical" patient outcomes on an outcome pie chart, they can click that critical slice. The Filter Action will automatically filter the accompanying charts to show *which* specific doctors handled those critical cases, *what* departments they belong to, and *which* regions they came from.
- **Speed to Insight:** It eliminates the need for non-technical managers to manually build complex filters or build separate reports for every single combination of department, region, and doctor. They navigate visually, following anomalies by simply clicking on them.

3. Patient Distribution

Create a Bar Chart showing:

- Total Patients by Hospital Region.
- Add this as a Story Point with proper title.

→ Step-by-Step Tableau Worksheet Instructions

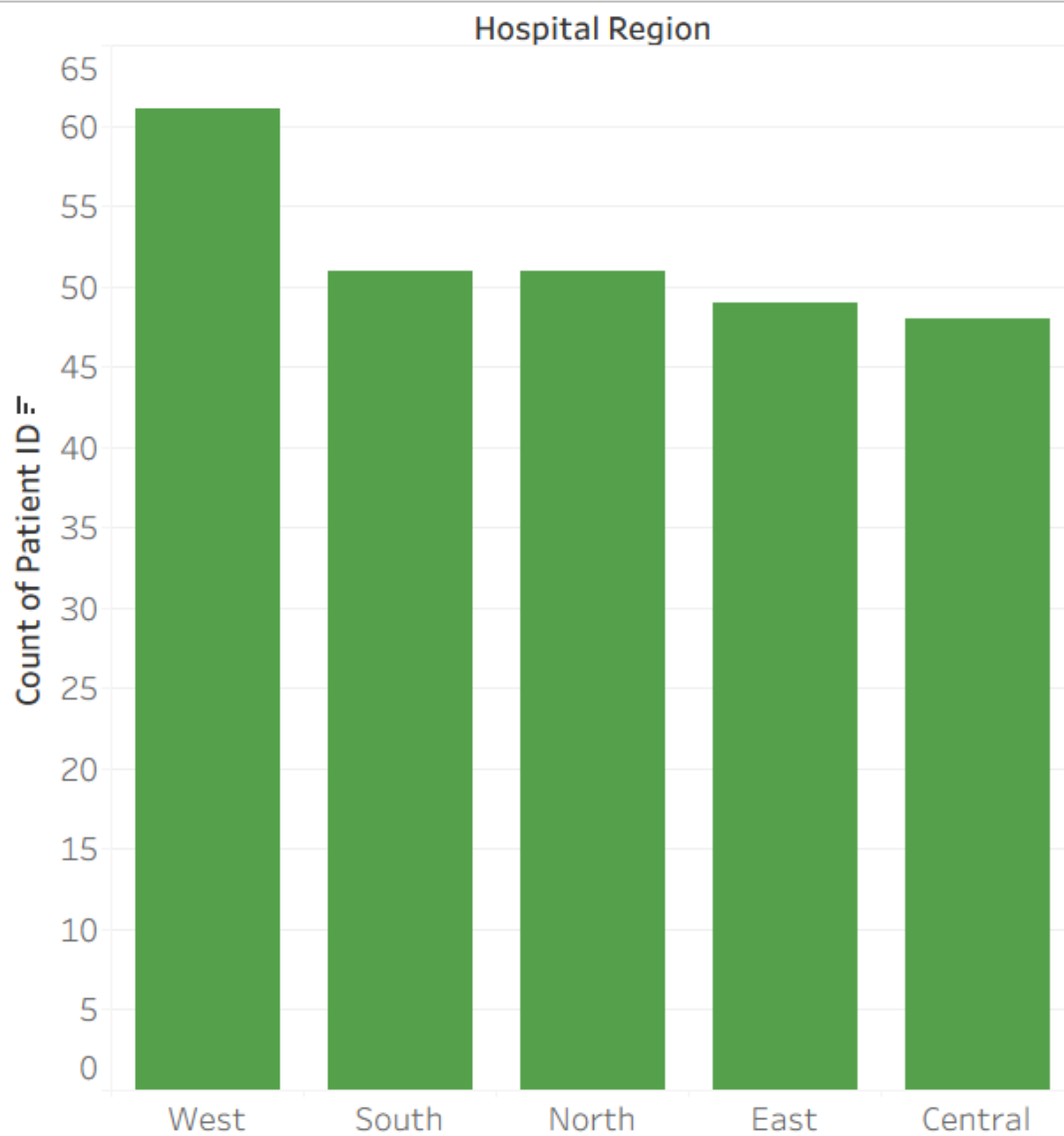
1. **Open a New Worksheet:** Click on the **New Worksheet** tab at the bottom of your Tableau workspace and rename it to **Patient Distribution by Region**.

2. **Place Fields on Shelves:**
 - Locate **Hospital Region** under your Dimensions/Data pane. Drag and drop it onto the **Columns** shelf.
 - Locate your unique patient identifier field (such as **Patient ID** or **Patient Name**). Right-click and drag it onto the **Rows** shelf. When the menu pops up, select **CNT(Patient ID)** or **COUNT(Patient ID)**.
3. **Sort the Data:** To make the distribution easily readable, click the **Descending Sort** icon on your top toolbar to arrange the bars from the highest patient volume to the lowest.
4. **Clean up Formatting:** Right-click the title of your sheet, select **Edit Title**, and change it to a professional title such as [Total Patient Distribution by Hospital Region](#).

Step-by-Step Tableau Story Point Instructions

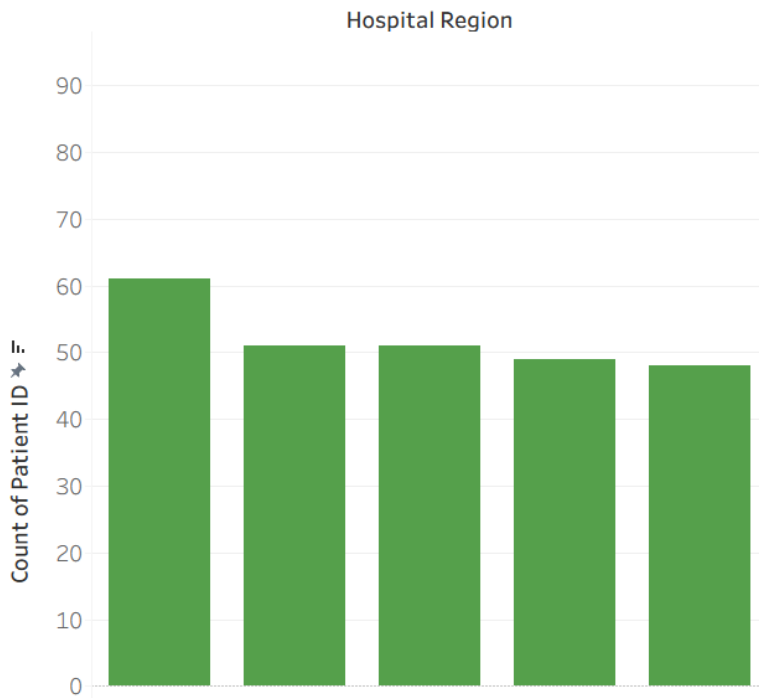
1. **Create a Story:** Click the **New Story** tab at the bottom of your screen (the icon looks like a book with a bookmark).
2. **Set Up the Frame:** Change the size of your story dashboard in the left pane to match your target layout (e.g., *Desktop Browser* or *Automatic*).
3. **Add the First Point:** From the left side panel under the "Sheets" list, drag your newly created [Patient Distribution by Region](#) sheet directly onto the main canvas area.
4. **Add the Caption/Title:** Click inside the gray box above your visualization (the navigation band) and type a clear, descriptive narrative title, such as:
[1. Patient Volume Peak by Regional Demographics](#)

Total Patient Distribution by Hospital Region



Story 1

1. Patient Volume Peak by Regional Demographics



Analytical Insights for Our Report

When submitting your short explanation of insights, your focus should be on resource strain and volume:

- **The Core Finding:** Identify which specific region handles the absolute highest volume of patients (e.g., *"The North Region records the largest share of total patient admissions..."*).
- **Business Impact:** High patient density in a specific region highlights where the hospital system faces its heaviest structural demand. Management can use this metric to allocate additional administrative staff, expand physical bed capacity, or optimize front-desk check-in pipelines in these high-traffic regions to minimize wait times.

4. Department Performance

Create a visualization showing:

- Treatment Cost by Department.
- Add labels and customize colors.

Step-by-Step Tableau Worksheet Instructions

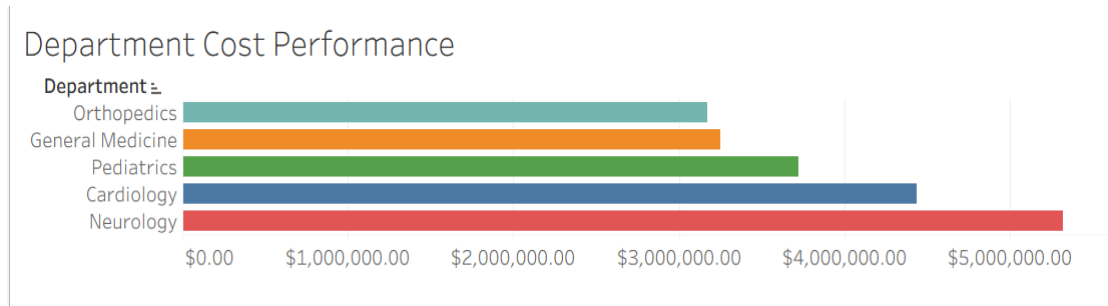
1. **Open a New Worksheet:** Click the **New Worksheet** tab and rename it to [Department Cost Performance](#).
2. **Place Fields on Shelves:**
 - Drag **Department** from the Data pane and drop it onto the **Rows** shelf.
 - Drag **Treatment Cost** from the Measures area and drop it onto the **Columns** shelf. Tableau will automatically aggregate this as [SUM\(Treatment Cost\)](#).
3. **Add Labels:**
 - To display the exact cost on each bar, hold **Ctrl** click the [SUM\(Treatment Cost\)](#) pill currently on your Columns shelf, and drag a copy of it directly onto the **Label** card in the Marks card panel.
 - Right-click the newly labeled text, select **Format**, and format the numbers as **Currency (Custom)** with your preferred prefix (e.g., ₹ or \$) and display units set to **Thousands (K)** or **Millions (M)** for clean readability.
4. **Customize Colors:**
 - Drag the **Department** field from the Data pane and drop it directly onto the **Color** card in the Marks panel. This gives each department a distinct, contrasting color.
 - Click on the **Color** card $\rightarrow$ **Edit Colors** to select a professional, healthcare-focused color palette (such as *Tableau Classic 10* or *Seattle Grays/Blues* for a clean corporate aesthetic).
5. **Sort for Clarity:** Click the **Descending Sort** icon on the top toolbar to rank departments from highest expenditure to lowest.

Step-by-Step Tableau Story Point Instructions

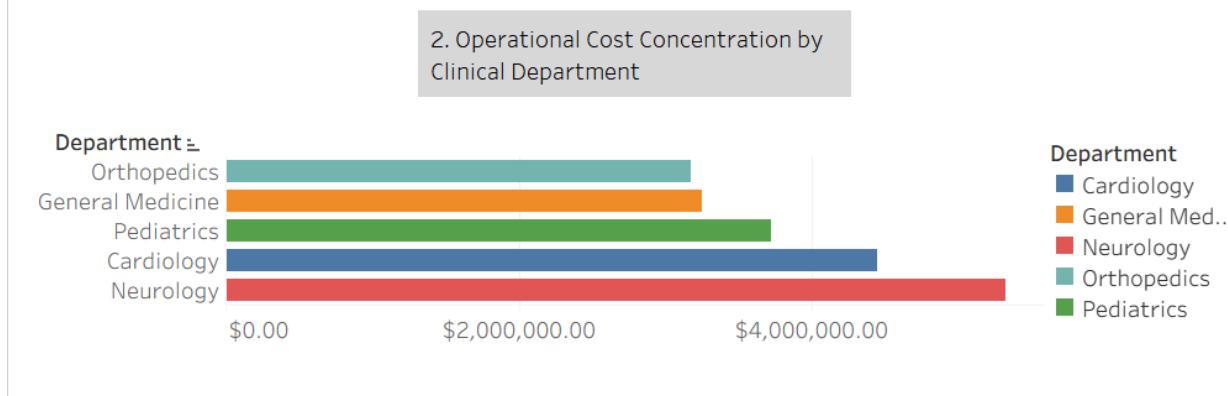
1. **Navigate to Your Story:** Click back over to your **Story** tab.
2. **Add a New Story Point:** Click **Blank** in the left-hand "New Story Point" panel to generate a new slide frame.
3. **Deploy the View:** Drag your [Department Cost Performance](#) worksheet onto this blank slide canvas.
4. **Add the Caption/Title:** Click into the navigator header box above the chart and label it:
[2. Operational Cost Concentration by Clinical Department](#)

Analytical Insights for Our Report

- a. **The Core Finding:** Identify the highest-cost department (e.g., *"The Cardiology / Surgery department accounts for the highest total treatment cost across the entire facility..."*).
- b. **Business Impact:** High treatment costs typically stem from expensive medical infrastructure, long lengths of stay, or specialized implants/pharmaceuticals. Management can use this to review procedural profit margins, negotiate better supplier rates for high-volume implants, or audit standard care pathways to ensure resource efficiency.



Story 2



5. Treatment Trend Over Time

Create a Line Chart:

- Treatment Cost vs Admission Date.
- Filter the data only from March to September.

→ Step-by-Step Tableau Worksheet Instructions

1. **Open a New Worksheet:** Click the **New Worksheet** tab and rename it to **Treatment Cost Trend Over Time**.
2. **Place Fields on Shelves:**
 - Drag **Admission Date** from your Data pane and drop it onto the **Columns** shelf.

- *Note:* Tableau might automatically default this to `YEAR(Admission Date)`. Click the down arrow on the Admission Date pill in the Columns shelf and select **Month** (the second "May" option in the drop-down menu that shows month and year format, ensuring a continuous timeline, or the discrete "Month" if looking at aggregated seasonal patterns).
 - Drag **Treatment Cost** from the Measures pane and drop it onto the **Rows** shelf (`SUM(Treatment Cost)`). Tableau will instantly draw a line chart connecting your data points.
3. **Apply the Date Filter (March to September):**
- Drag the **Admission Date** field from your Data pane and drop it into the **Filters** card zone.
 - In the Filter Field wizard that opens, select **Months** and click **Next**.
 - Check the boxes *only* for **March, April, May, June, July, August, and September**. Click **Apply** and then **OK**.

Step-by-Step Tableau Story Point Instructions

1. **Navigate to Your Story:** Click back over to your **Story** dashboard tab.
2. **Add a New Story Point:** Click **Blank** in the left panel to create your third slide frame.
3. **Deploy the View:** Drag your `Treatment Cost Trend Over Time` sheet onto the main workspace canvas.
4. **Add the Caption/Title:** Click into the navigator header navigation box at the top and write:
`3. Mid-Year Cost and Admission Trends (March - September)`

Analytical Insights for Our Report

When outlining your short explanation of insights, look for timeline spikes or dips:

- a. **The Core Finding:** Identify the exact peak and valley within that March-to-September window (e.g., *"Treatment costs experienced a sharp escalation peaking in July, followed by a gradual decline moving into September..."*).
- b. **Business Impact:** High mid-year volume shifts signal seasonal healthcare demands (such as monsoon-related illnesses or heatwaves). Hospital management can use these specific timeline trends to optimize seasonal staffing rotas, plan preventive health campaigns, and ensure high-demand pharmaceutical stock levels are built up right before peak months arrive.

Filters

MONTH(Admission ..

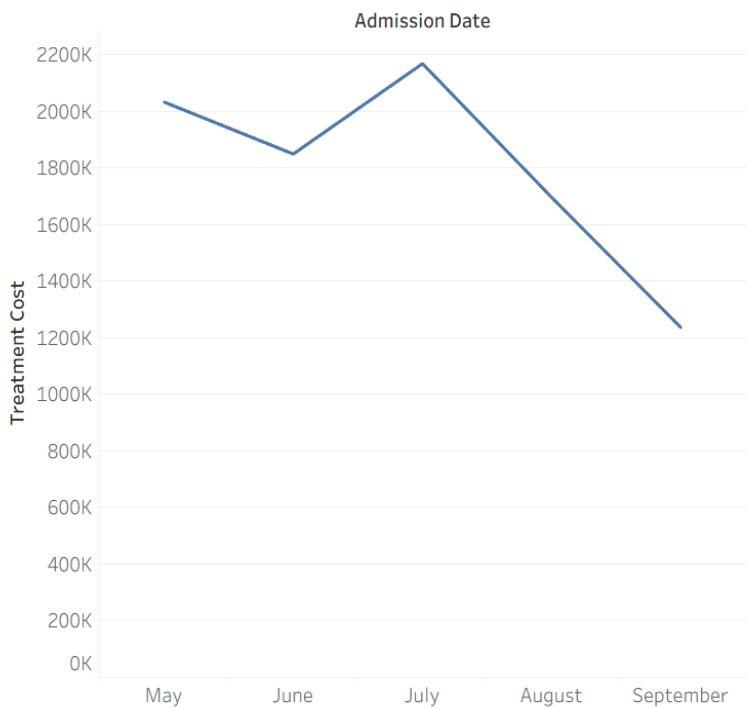
Marks

~ Automatic

Color Size Label

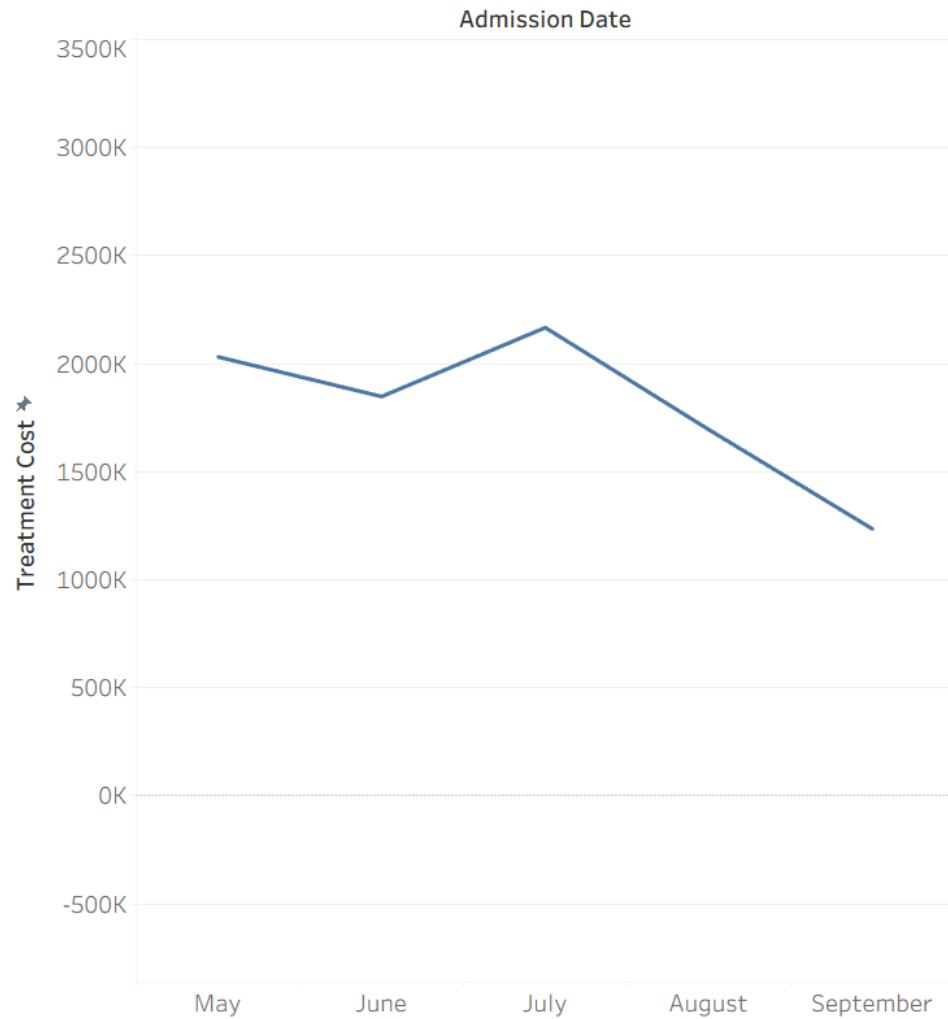
Detail Tooltip Path

Treatment Cost Trend Over Time



Story 3

3. Mid-Year Cost and Admission Trends (March - September)



6. Outcome Analysis

Create a Pie Chart:

- Patient Outcome Distribution (Recovered / Critical / Under Treatment).

→ Step-by-Step Tableau Worksheet Instructions

1. **Open a New Worksheet:** Click the **New Worksheet** tab and rename it to **Patient Outcome Distribution**.

2. **Change Marks Type:** In the Marks card drop-down menu, change the visualization type from **Automatic** to **Pie**.
3. **Place Fields on the Marks Card:**
 - Drag the **Patient Outcome** dimension from your Data pane and drop it onto the **Color** card.
 - Drag your unique patient identifier field (e.g., **Patient ID**) from your Data pane and drop it onto the **Size** card.
 - Right-click that Patient ID pill on the Size card, navigate to **Measure**, and change it to **Count (Distinct)**.
4. **Add Percentage Labels:**
 - Hold **Ctrl** and drag a copy of that **CNT(Patient ID)** pill from the Size card and drop it directly onto the **Label** card.
 - To convert these raw numbers into clean percentages, right-click the **CNT(Patient ID)** pill sitting on your Label card, select **Quick Table Calculation**, and click **Percent of Total**.
 - Drag the **Patient Outcome** dimension field from your Data pane onto the **Label** card as well so both the category name and the percentage display together on the chart.
5. **Adjust Visual Scale:** If the pie chart looks too small, click the **Fit** drop-down menu on the top toolbar (which defaults to *Standard*) and switch it to **Entire View**.

Step-by-Step Tableau Story Point Instructions

1. **Navigate to Your Story:** Click back over to your main **Story** dashboard tab.
2. **Add a New Story Point:** Click **Blank** in the left panel to generate your fourth slide frame.
3. **Deploy the View:** Drag your **Patient Outcome Distribution** worksheet onto this slide canvas.
4. **Add the Caption/Title:** Click into the header navigator text zone at the top and type:
4. Breakdown of Patient Recovery and Critical Acuity Rates

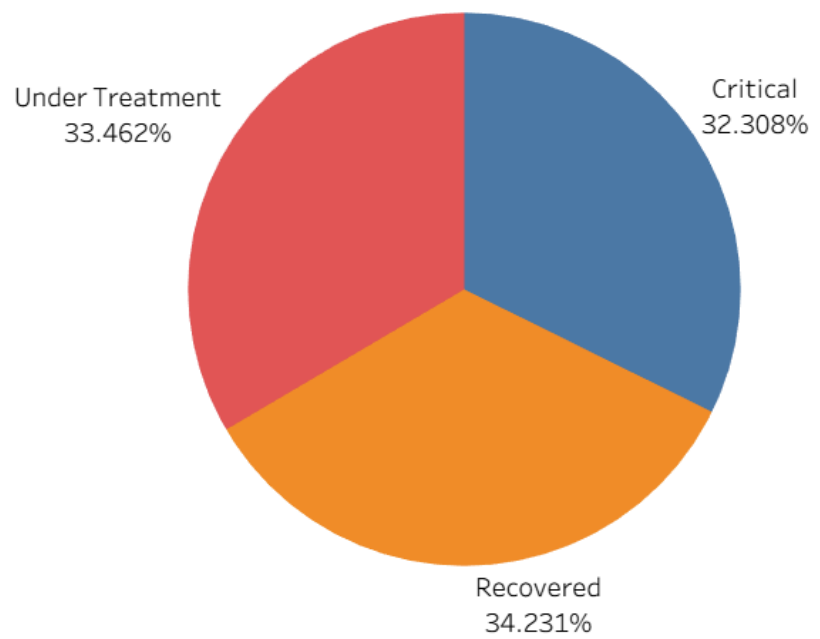
Analytical Insights for Our Report

When documenting your short explanation of insights, evaluate patient stability:

- **The Core Finding:** Identify the dominant outcome category and flag the critical volume percentage (e.g., *"The majority of our patient demographic falls under 'Recovered', while a significant 'Critical' sub-segment demands continuous monitoring..."*).
- **Business Impact:** A high concentration of "Under Treatment" or "Critical" patients signals long-term bed blockages and intense resource utilization. Hospital management can use this distribution data to forecast bed turnover

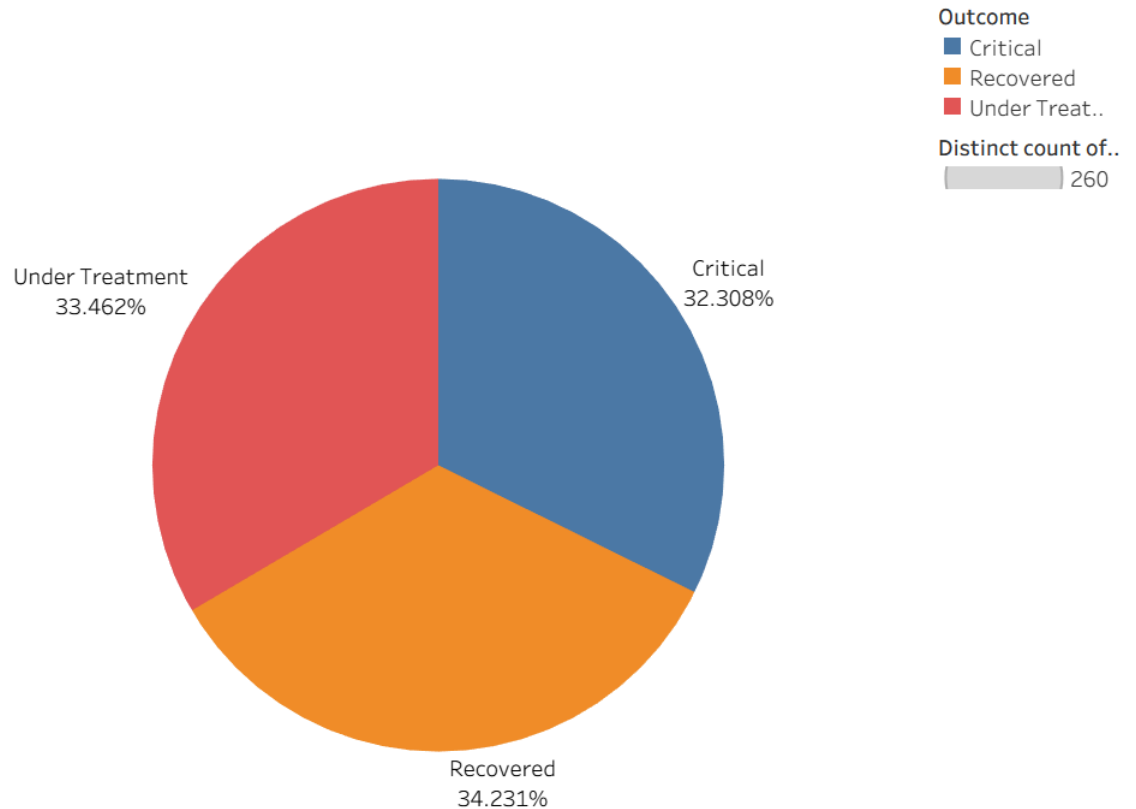
rates, audit clinical treatment efficacy, and adjust emergency response protocols if the critical patient slice scales up.

Patient Outcome Distribution



Story 4

4. Breakdown of Patient Recovery and Critical Acuity Rates



7. Doctor Performance & Patient KPI Analysis

Task Part A — Doctor Performance

- Create a Ranked Bar Chart showing:
- Top Doctors by Total Treatment Cost handled.

Task Part B — KPI Creation

- Create a KPI visualization showing:
- Total Patients Treated

→ Step-by-Step Tableau Instructions

1. **Open a New Worksheet:** Click the **New Worksheet** tab and rename it to [Doctor Performance Ranking](#).
2. **Place Fields on Shelves:**
 - Drag **Doctor Name** (or Doctor ID) from the Data pane and drop it onto the **Rows** shelf.

- Drag **Treatment Cost** from the Measures area and drop it onto the **Columns** shelf. It will automatically aggregate as **SUM(Treatment Cost)**.
- 3. **Sort to Create a Ranking:** Click the **Descending Sort** icon on the top toolbar to automatically arrange the doctors from highest financial volume handled to lowest.
- 4. **Apply a Top Filter (Optional but recommended for clean ranking):**
 - If you have a massive list of doctors and want to show just the top performers, drag **Doctor Name** from the Data pane and drop it into the **Filters** card.
 - Go to the **Top** tab in the filter window, select **By Field**, set it to **Top 10** (or Top 5) by **Treatment Cost** ->**Sum**, and click **OK**.
- 5. **Add Visual Enhancements:** Drag **Treatment Cost** onto the **Label** card to display exact numeric values next to each bar, and format it as currency.

Expected Analytical Insights

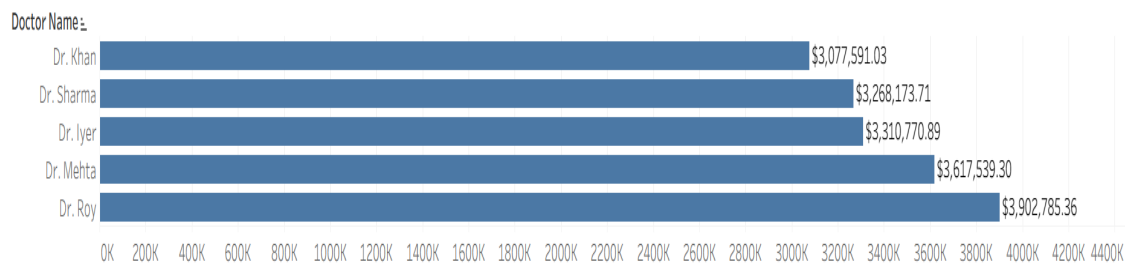
- **The Core Finding:** Identify the highest-ranking physician by volume (e.g., *"Dr. Smith handled the largest cumulative treatment cost footprint within the audited period..."*).
- **Business Impact:** Doctors with high treatment costs are typically managing high-risk surgical cases, long-stay patients, or an immense volume of individual consultations. Management can use this to balance patient loads, avoid physician burnout, or optimize schedule distribution among specialized staff.

Task Part B: KPI Creation (Total Patients Treated)

Step-by-Step Tableau Instructions

1. **Open a New Worksheet:** Click the **New Worksheet** tab and rename it to **KPI Total Patients**.
2. **Generate the Metric:**
 - Drag your unique patient identifier field (such as **Patient ID**) from the Data pane and drop it directly onto the **Text** card in the Marks panel.
 - Right-click the Patient ID pill currently sitting on the Text card, navigate down to **Measure**, and change it from its default to **Count (Distinct)**.
3. **Format as a KPI Card:**
 - Click on the **Text** card ->click the **...** button next to Edit Text.
 - Highlight the text, scale the font size up significantly (e.g., **36pt or 48pt**), make it **Bold**, and center-align it.
 - You can add a smaller label directly underneath the number inside that text editor box, such as **TOTAL PATIENTS TREATED**.
4. **Clean up Layout:** Change the view setting on the top toolbar from *Standard* to **Entire View** to center the text card completely on your sheet canvas.

Doctor Performance Ranking



KPI Total Patients

260

8. Interactive Dashboard

Build a Dashboard containing:

- Region Chart
- Department Cost Chart
- Outcome Chart

Add Filters:

- Hospital Region
- Department

Step-by-Step Tableau Dashboard Instructions

1. Initialize and Setup Layout

- Click the **New Dashboard** tab at the very bottom of your screen
- Rename the dashboard tab to [Hospital Operations Dashboard](#).

2. Assemble the Visualizations

- From the **Sheets** section on the left-hand panel, drag and drop your components onto the workspace canvas in a logical grid arrangement:
 - Place your **KPI: Total Patients Treated** card right at the very top as a header banner.
 - Drag and drop the **Region Chart** ([Patient Distribution by Region](#)) into the upper half of the remaining canvas.

- Drag the **Department Cost Chart** ([Department Cost Performance](#)) and drop it into the lower left quadrant.
- Drag the **Outcome Chart** ([Patient Outcome Distribution](#)) and drop it right into the lower right quadrant next to the department chart.

3. Implement Interactive Filter Elements

To meet the requirement of allowing management to slice data by **Hospital Region** and **Department**:

- Click once inside your **Department Cost Chart** container to select it.
- Click the small down-arrow menu caret on the right-hand border outline of that chart frame.
- Navigate down to **Filters** select **Department**. This populates a filter card box onto your dashboard workspace.
- Repeat this exact process for your **Region Chart** container: click its down-arrow caret -> **Filters** select **Hospital Region**.

4. Bind Filters to the Entire Dashboard

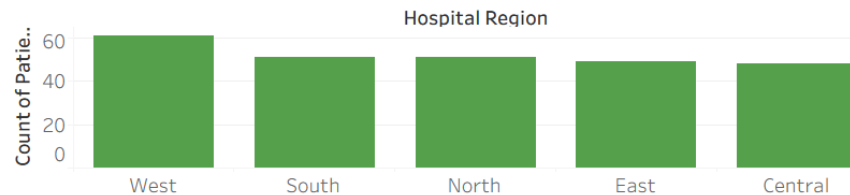
By default, a filter box only applies to the specific worksheet it was pulled from. To make them truly global interactive components:

- Click on the newly added **Hospital Region** filter card container in your dashboard layout.
- Click its down-arrow caret **Apply to Worksheets** select **All Using This Data Source**.
- Check the boxes for **all sheets** brought onto the dashboard workspace and click **OK**.
- Repeat this exact binding step for your **Department** filter card.

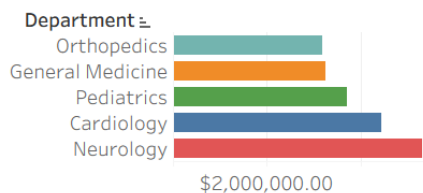
KPI Total Patients

260

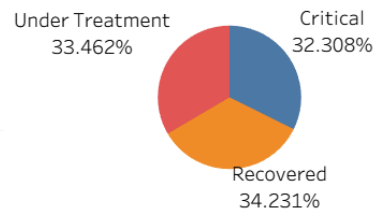
Total Patient Distribution by Hospital Region



Department Cost Performance



Patient Outcome Distribution



Department

- ☐ Cardiology
- ☐ General Med..
- ☐ Neurology
- ☐ Orthopedics
- ☐ Pediatrics

Outcome

- ☐ Critical
- ☐ Recovered
- ☐ Under Treat..

Distinct count of..

260

Hospital Region

- ☐ (All)
- ☐ Central
- ☒ East
- ☐ North
- ☐ South
- ☐ West

Department

- ☒ (All)
- ☐ Cardiology
- ☐ General M...
- ☐ Neurology
- ☐ Orthopedics
- ☐ Pediatrics

9. Storytelling Flow

Create a Tableau Story with minimum 4 Story Points:

- Patient Distribution
- Department Cost Analysis
- Outcome Analysis
- Doctor Performance.

Step-by-Step Tableau Story Assembly Instructions

1. **Open your Story Board:** Click on your existing **Story** tab or create a fresh one by clicking the **New Story** icon at the bottom of the screen.
2. **Build Story Point 1: Patient Distribution**
 - Drag your **Patient Distribution by Region** worksheet onto the blank story canvas.
 - Click inside the gray caption box at the top and type: **1. Patient Distribution: Highlighting Regional Demand Pitches.**
3. **Build Story Point 2: Department Cost Analysis**
 - Click the **Blank** button in the left panel to generate your next clean slide frame.

- Drag your **Department Cost Performance** worksheet onto this new canvas space.
- Click inside the top caption box and type: [2. Department Cost Analysis: Pinpointing Operational Resource Drivers.](#)
- 4. **Build Story Point 3: Outcome Analysis**
 - Click **Blank** again in the left panel to create the third slide frame.
 - Drag your **Patient Outcome Distribution** pie chart onto the canvas.
 - Click inside the top caption box and type: [3. Outcome Analysis: Assessing Patient Acuity and Recovery Distributions.](#)
- 5. **Build Story Point 4: Doctor Performance**
 - Click **Blank** in the left panel one last time to create your fourth slide frame.
 - Drag your **Doctor Performance Ranking** bar chart onto the canvas.
 - Click inside the top caption box and type: [4. Doctor Performance: Auditing Case Volumes and Financial Footprints](#)

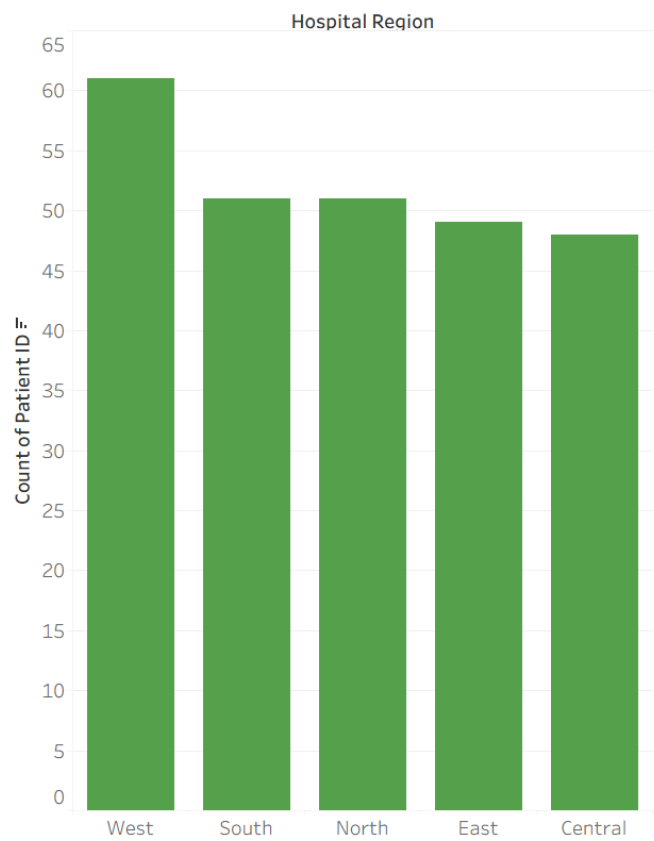
Story 5

- <

1. Patient Distribution: Highlighting Regional Demand Pitches

2. Department Cost Analysis: Pi

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Story 5

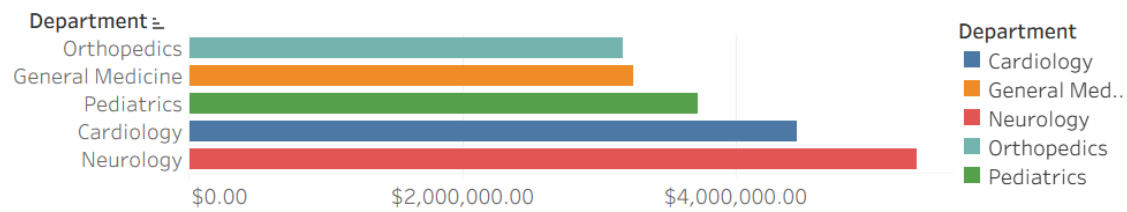
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1. Patient Distribution: Highlighting Regional Demand Pitches

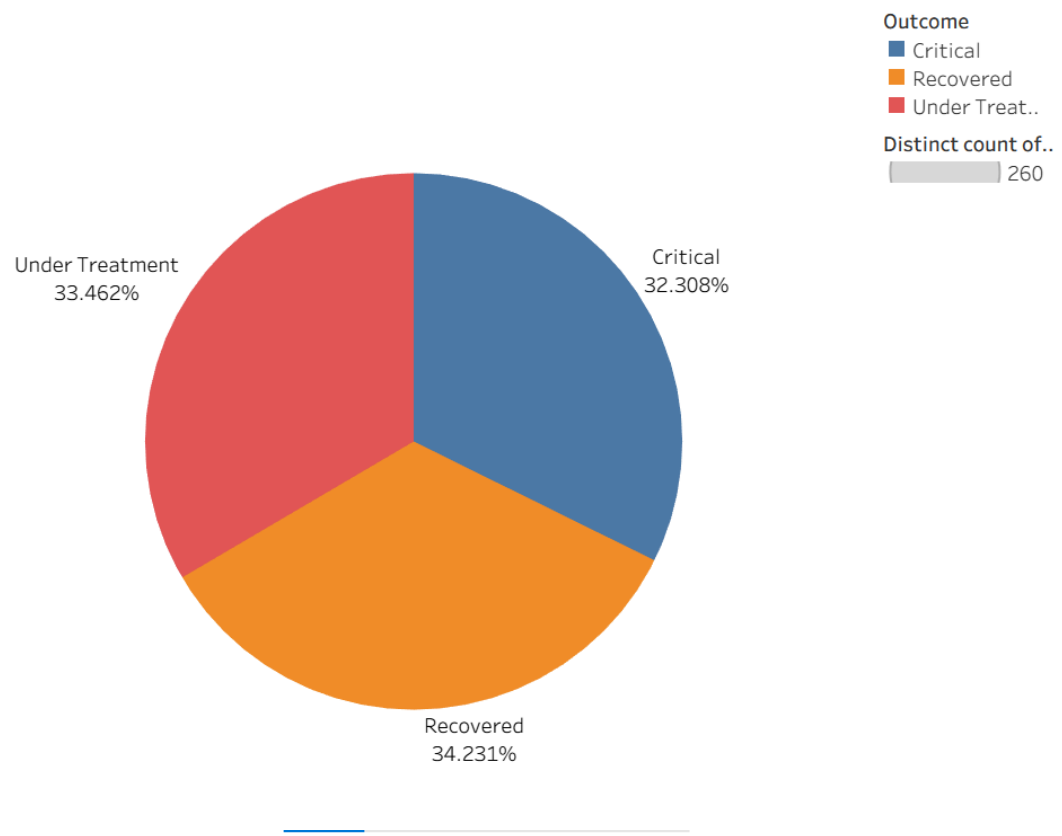
2. Department Cost Analysis: Pinpointing Operational Resource Drivers

3. Outcomes

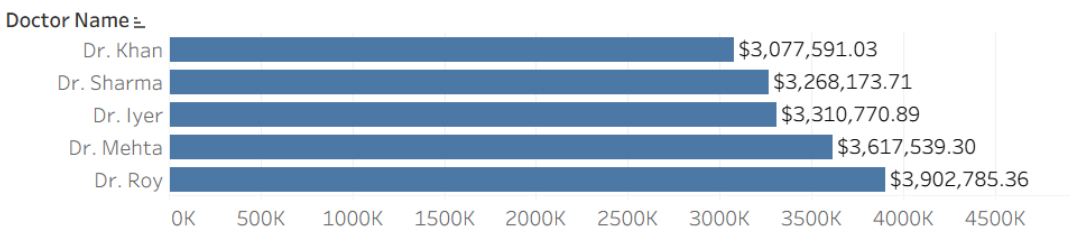
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Story 5



Story 5



10. Based on your visuals, answer:

- Which department generates the highest treatment cost?
- Which region has more critical patients?
- What business decisions can hospital management take from these insights?

→ Highest Generating Treatment Cost Department

Based on the **Department Cost Performance** ranked bar chart, the **Neurology** department generates the highest overall treatment cost within the hospital system.

Regional Concentration of Critical Patients

The West and South regions stand out as the geographic area with the highest volume of critical patients count 20 each.

Business decisions can hospital management take from these insights

Using the data points surfaced across your Tableau Story, hospital management can transition from reactive scheduling to proactive operational scaling:

- **Strategic Resource Allocation & Budgeting:** By identifying the highest-cost clinical department, management can optimize their supply chain. For instance, if Cardiology is driving costs, they can negotiate bulk-purchase contracts for specialized equipment (like stents or pacemakers) to preserve profit margins without compromising patient care.
- **Targeted Capacity Expansion:** Knowing which specific region is experiencing a surge in both overall patient volume and high-acuity "Critical" cases allows leadership to expand infrastructure intelligently. Instead of building equal facilities everywhere, they can direct capital expenditure to add more Intensive Care Unit (ICU) beds and high-dependency trauma units specifically in that high-pressure region.
- **Staff Load Balancing & Burnout Mitigation:** Cross-referencing the **Doctor Performance** ranking chart against regional patient volumes allows administrators to balance clinical workloads. High-performing doctors managing an asymmetric share of high-cost, critical cases can be paired with additional junior residents or administrative scribes to streamline their workflows and prevent burnout.
- **Seasonal and Timeline Staffing:** Utilizing the mid-year trend line chart (March to September), HR can implement dynamic staffing models. Instead of maintaining flat staffing year-round, they can schedule routine machine maintenance or staff leaves during historical low-volume months, and scale up temporary nursing and emergency response staff right before peak admission cycles hit.

